This full-text paper was peer-reviewed at the direction of IEEE Instrumentation and Measurement Society prior to the acceptance and publication.

2025 IEEE International Instrumentation and Measurement Technology Conference (I2MTC) | 979-8-3315-0500-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/I2MTC62753.2025.11079100

Cellular Signals-of-Opportunity for RF Propagation Measurements with SDRs

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***Abstract*—Cell towers broadcast reference signals that mobile phones measure and report as reference signal received power (RSRP). Phones use the reference signal measurements for connectivity and handover between towers. This study explores the use of software defined radios (SDRs) for precise measurement of the power levels of these reference signals, showing them to be a rich yet untapped source for high-quality radio frequency (RF) propagation measurements. Given the widespread availability of cell towers, these signals are well-suited to generating the extensive RF channel data required by deep learning and ray-tracing models. However, questions remain about whether SDRs can achieve the level of accuracy RF researchers need for propagation model validation and enhancement.**

We present sdrRSRP, an open-source SDR toolset offering comprehensive, low-level control over reference signal measurements. This study shows that, with proper calibration, an SDR can deliver reference signal power measurements with an accuracy of ± 3.5 dB. Additionally, we compare reference signal measurements to the continuous-waveform measurement technique commonly used by RF propagation engineers, demonstrating that the two methods yield measurements with negligible differences (< 1 dB).

***Index Terms*—Radio propagation, SDR, signals-of-opportunity.**

I. Introduction

High-quality radio frequency (RF) propagation measurements are essential for the wireless community. In 2022, the Federal Communications Commission (FCC) Technical Advisory Council emphasized the need for real-world propagation data to validate emerging RF propagation models [1]. These models are crucial for addressing challenges such as spectrum sharing, expanding rural coverage, and mitigating interference in crowded RF environments.

As neural network–based propagation models and ray-tracing models gain traction [2]–[4], demand has surged for propagation data to meet the needs of model training and calibration. The Commerce Spectrum Management Advisory Committee in its December 2023 recommendations to the National Telecommunication and Information Administration (NTIA) identified these models as integral to the future of spectrum management and policy formulation [5].

Unfortunately, the collection of quality RF data, used for the creation and validation of propagation models, is resource-intensive [6]. It typically requires laboratory-grade measurement equipment, a skilled engineering team, and, most critically, a federal experimental license to transmit an RF signal at the required frequency and power levels [7].

To address these significant barriers, this study investigates the use of cellular reference signals as a signal-of-opportunity to collect quality RF propagation data. *Cell-specific reference signals* (CRSs) from 4G networks are used by phones for cell handover. Phones average CRS power into a metric called reference signal received power (RSRP). A single cell tower broadcasts hundreds of CRSs per millisecond, and each offers an opportunity to capture extensive propagation data at various mid-band frequencies (500 MHz to 6 GHz) across urban and rural settings. Since cell towers also broadcast the CRS transmission power, a complete propagation loss calculation can be computed. Despite these benefits, the use of CRS measurements has not been seriously considered by the RF community as a source for propagation measurements.

Three key challenges have limited the adoption of CRS measurements for RF data collection: (1) the exact RSRP averaging algorithm on phones is proprietary, and it sparsely samples the CRSs; (2) traditional spectrum analyzers do not measure and report the CRS values from the 4G downlink; and (3) it has yet to be shown that software-defined radios (SDRs) can measure and produce accurate CRS powers.

This is the first study to develop and validate an SDR-based tool for CRS measurements. We present `sdrRSRP`, an open-source SDR toolset offering comprehensive, low-level control over CRS measurements. Through the description and results of an extensive laboratory and real-world measurement campaign, this study shows that with proper calibration an SDR can deliver, with an accuracy of ± 3.5 dB, mobile propagation measurements of CRSs from any commercial 4G tower. Additionally, we compare the wideband CRS measurements to the narrowband continuous-waveform (CW) measurement technique commonly used by RF propagation engineers, demonstrating that the two methods yield measurements with negligible differences (< 1 dB). We limited the scope of the experiments in this study to 4G reference signals and intend to study of 5G signals in future work.

II. Background

This section describes how the CRSs are broadcast by cell towers and how the signal processing is performed by `sdrRSRP` to measure them. This process is specified in 3GPP Technical Specification (TS) 36.213 [8].

This work was funded in part by NSF SpectrumX Center, Award #2132700 and by the Institute for Telecommunication Sciences, National Telecommunications and Information Administration, via the U.S. Office of Personnel Management Pathways (internship) Program.

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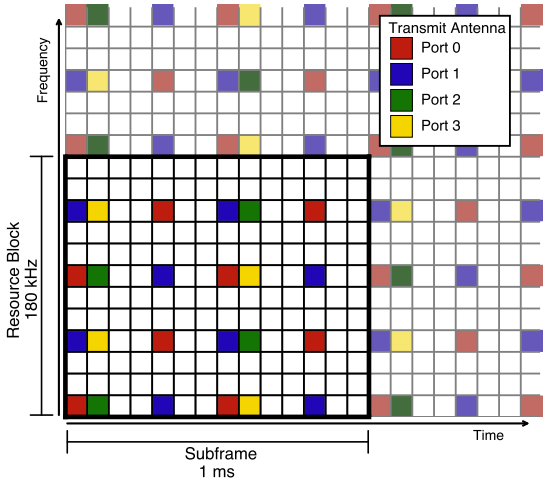

Fig. 1. The time and frequency mapping of reference signals within the 4G resource grid (downlink). The red, blue, green, and yellow elements carry the reference signals for the four possible transmit antennas.

The 4G signal uses orthogonal frequency-division multiplexing (OFDM) modulation, meaning that transmitted CRSs are interleaved in both time and frequency across the entire downlink channel. To accurately extract each individual reference signal power, the SDR must perform proper demodulation of the multiplexed signals.

Figure 1 illustrates a time and frequency arrangement of the CRSs in the 4G resource grid. The bold-bordered rectangle is one subframe in time and one resource block (RB) in frequency. A subframe is one millisecond long and the RB is 180 kHz wide: it contains 12×14 of the smaller resource elements. The colored (red, blue, green, or yellow) resource elements carry a CRS and indicate the transmitting antenna port. The CRS locations repeat every subframe and RB.

The CRS measurement begins with the SDR sampling the downlink center frequency at 1.92 MS/s (Megasamples per second) to collect the central six RBs of the downlink signal. These RBs contain the Primary and Secondary Synchronization Signals (PSS and SSS). The SDR performs successive PSS and SSS auto-correlations until a correlation peak is detected and timing synchronization is complete.

The PSS and SSS are used to compute a cell tower's Physical Cell ID (PCI), and time synchronization allows the SDR to decode the Master Information Block (MIB) containing three configuration settings the SDR needs to decode the CRSs: the transmission mode of Frequency-Division Duplexing (FDD) or Time-Division Duplexing (TDD), downlink bandwidth, and number of transmit antenna ports (1, 2, or 4).

The SDR reconfigures its sampling rate based on the bandwidth. For example, an SDR measuring a 10 MHz (50 RBs) downlink samples at 11.52 MS/s. One subframe is acquired at a time. Successive discrete Fourier transforms (DFT) are performed on the subframe. The DFT size is also based on the bandwidth, with the resulting resolution bandwidth always being 15 kHz, the spacing of an individual subcarrier.

Continuing with the 10 MHz downlink example, in one subframe the antenna ports 0 and 1 each transmit 400 CRSs and antenna ports 2 and 3 each transmit 200 CRSs, for a total of 1200 CRSs [8]. The 3GPP defines the per-subframe RSRP as the linear average over the power contributions (in Watts) of the 1200 CRSs in a subframe [9].

TABLE I
MOBILE PHONE REFERENCE SIGNAL MEASUREMENT LENGTHS AND PERIODICITY, AS SPECIFIED BY 3GPP [10]

Measurement Length (ms)	Repetition Period (ms)	Percent of Reference Signals Measured
6	40	15%
6	80	8%
6	160	4%
3	40	8%
3	80	4%
3	160	2%

To report the per-subframe RSRP, the average of the CRS power is calculated,

$$\text{RSRP} = \frac{1}{N}\sum_{i=1}^{N} |\text{CRS}_i|^2$$

This value is converted to dB and then calibrated to convert to dBm based on the gain setting G within the SDR,

$$\text{RSRP}_{\text{dBm}} = 10\log_{10}(\text{RSRP}) + G$$

We use `sdrRSRP` to report power from the thousands of CRSs per millisecond. By contrast, mobile phones can only measure CRSs intermittently and can report but one RSRP per second. Table I shows 3GPP-specified measurement lengths and repetition periods for the intermittent CRS measurements. This table highlights two drawbacks of phone RSRP measurements: (1) the sampling and averaging algorithm varies and is unique to the phone model, and (2) only a small percentage of the CRSs are actually measured in a phone's RSRP calculation, as shown in the percent of reference signals measured column.

III. Related Work

There are a number of mobile phone applications and dedicated hardware devices for the measurement of RSRP and other cellular metrics (e.g., Received Signal Strength Indicator and Reference Signal Received Quality). There are two phone applications for collecting cellular metrics: SigCap [11] and G-NetTrack Lite/Pro [12]. They run on standard (not rooted) Android phones (since iOS prohibits RSRP reporting an application) and produce one RSRP per second with a resolution of one dB.

The accuracy of RSRP measurements from phones have been evaluated for use in in-building network coverage [13] and propagation modeling [14]. Cellular metric data from phones have been collected to study the pace of the ongoing commercial 5G rollout [15]. There also exists literature on fabricating dedicated hardware devices with custom 4G chipsets to collect physical layer measurements [16], [17].

MobileInsight [18] is a tool for collecting cellular metrics from the chipset of a rooted phone. We tested its mobile app

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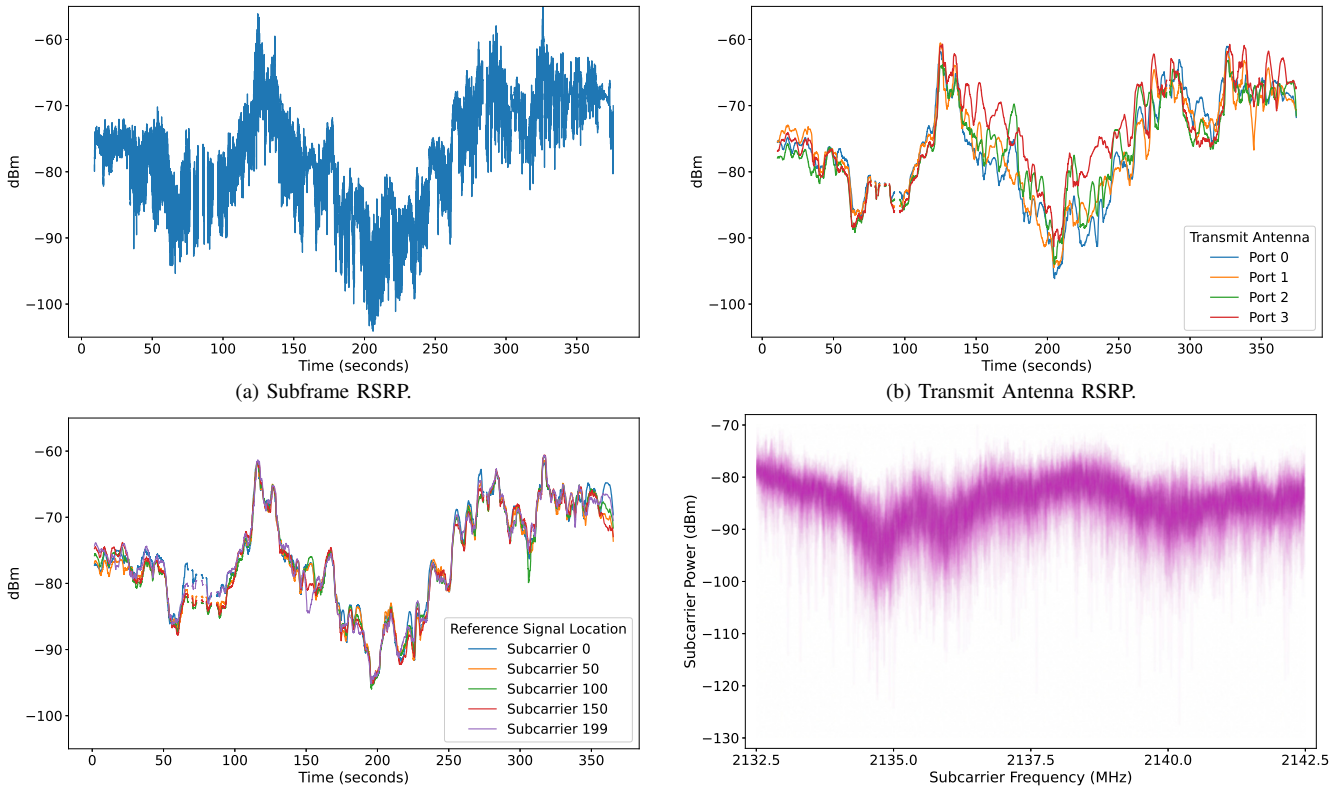

(a) Subframe RSRP.

(b) Transmit Antenna RSRP.

(c) Individual Subcarrier RSRP.

(d) Channel State Coefficient RSRP.

Fig. 2. Measurements from a field trial with the `sdrRSRP`, illustrating the different RSRP measurement modes. (a) Subframe RSRP (top-left), (b) Transmit Antenna RSRP (top-right), (c) Individual Subcarrier RSRP (bottom-left), (d) Channel State Coefficient RSRP (bottom-right).

(v6.0.0) with the NetLogger plugin on a rooted Huawei Nexus 6P and a rooted Google Pixel 3; it produced one RSRP sample per second. QualiPoc [19] is an operating system that runs on rooted Android phones. We also tested signal analyzers from Keysight [20], Rohde & Schwarz [21], and PCTEL [22][1].

All the tools reported above have the capability to report the RSRP metric, with the most common resolution being one RSRP sample-per-second. None of the above tools provide access to the raw reference signals at a per-millisecond and per-subcarrier level. And most importantly, the above tools are opaque, as they do not provide any insight into the algorithm used to compute the average RSRP from the raw reference signal power measurements.

IV. SDRRSRP Measurement Tool

To fully control of the CRS measurement system, we developed the `sdrRSRP` measurement tool, an open-source software tool for use with SDR hardware that can measure any 4G network. To develop `sdrRSRP`, we built onto srsUE [23], an open-source software suite of the mobile phone radio. `sdrRSRP` provides users low-level control of the CRS measurements, including measuring individual subcarrier reference signal power, selecting sampling intervals, and controlling RSRP averaging. We tested `sdrRSRP` on the Ettus USRP B210 and B205mini.

[1]Certain commercial products are identified in this report to specify adequately the technical aspects of the reported results. In no case does such identification imply recommendation or endorsement by NTIA, nor does it imply that the product identified is the best available for this purpose.

A. sdrRSRP Measurement Modes

`sdrRSRP` collects and reports the CRSs that a typical mobile phone intermittently captures but quickly discards. `sdrRSRP` has four distinct measurement modes, shown in Figure 2, which can be used individually or in tandem.

- *Subframe RSRP Mode* provides a per-subframe RSRP measurement. Each subframe is 1 ms long and the RSRP is the linear average of all CRSs within the subframe. This average includes the reference signals from all transmit antennas.
- *Transmit Antenna RSRP Mode* provides a per–transmit antenna port RSRP measurement every 1 ms. The CRSs are separated per transmit antenna port and are averaged and reported every subframe.
- *Individual Subcarrier RSRP Mode* provides the CRS power in subcarriers specified by the user. Each specified subcarrier is collected (without averaging) and reported for every subframe.
- *Channel State Coefficient RSRP Mode* provides the CRS power for all subcarriers across the band for a user-specified interval of time. At each interval, a new snapshot is created with channel state coefficients for the full bandwidth.

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TABLE II
SDRRSRP'S DYNAMIC RANGE FOR RSRP MEASUREMENTS RUNNING ON A CALIBRATED USRP B210 SDR

	Receive (Rx) Gain Setting		
	40 dB	50 dB	60 dB
Maximum RSRP	-40 dBm	-50 dBm	-60 dBm
Minimum RSRP	-90 dBm	-100 dBm	-105 dBm

Figure 2 illustrates data collected in each measurement mode during a field trial performed by `sdrRSRP`. This trial was a 400-second moving measurement of a 10 MHz bandwidth signal from an AT&T cell tower, covering 2 kilometers on the university campus. It was conducted with the same hardware setup described in the evaluation in Section IV-C.

The subframe RSRP plot (Fig. 2 (a)) shows the RSRP for every subframe and thus looks noisy, due to the fast-fading channel effects present in the signal. The transmit antenna and individual subcarrier plots (Fig. 2 (b-c)) show the RSRPs with a 1-second rolling average applied, which smooths out most of the fast-fading channel effects. The channel state coefficient plot (Fig. 2 (d)) shows the power for all CRSs across the 20 MHz band laid atop one another during a 100 ms capture, which explains the blurred effect.

Using `sdrRSRP` has the advantage of providing low-level control (e.g., control of the measurement bandwidth, time interval, and application of a rolling average) that is unavailable with other software and hardware tools. For example, where a mobile phone detects a 4G downlink as a single signal-of-opportunity, `sdrRSRP` sees hundreds of frequency and spatially distinct signals-of-opportunity transmitted by as many as four different antennas.

B. Calibration

Low-cost SDRs require calibration for accurate signal power measurements [24]. For those with access to laboratory-grade equipment, a high-quality calibration can be made with a spectrum analyzer used as the reference device. For those without access to laboratory-grade measurement equipment, we have found that `sdrRSRP` can be calibrated to the measurements of a mobile phone. Here we describe both methods.

For the laboratory-grade calibration, we created a cell tower in the lab using a second SDR (B210) running srsENB [23], an open-source 4G cell tower. The 4G downlink signal was transmitted across coaxial cables and a splitter to the `sdrRSRP` and the spectrum analyzer. Cable and splitter losses were accounted for. We set the tower's center frequency and transmission power and configured it to have 4 transmit antenna ports, 1 downlink control channel symbol, and a 10 MHz bandwidth. This configuration maximizes the number of CRSs and reduces the chance of interfering symbols.

With the spectrum analyzer, we found the average power in the highest 12 RBs (2.2 MHz) of the downlink. With no UEs attached to the tower, these RBs carry power only from the CRSs. A calibration factor (i.e., static linear shift) was calculated to correct the discrepancy between `sdrRSRP` and the spectrum analyzer. Each calibration factor is specific to the center frequency, e.g., a separate calibration factor must be found for band 2 (1960 MHz) and band 12 (737.5 MHz).

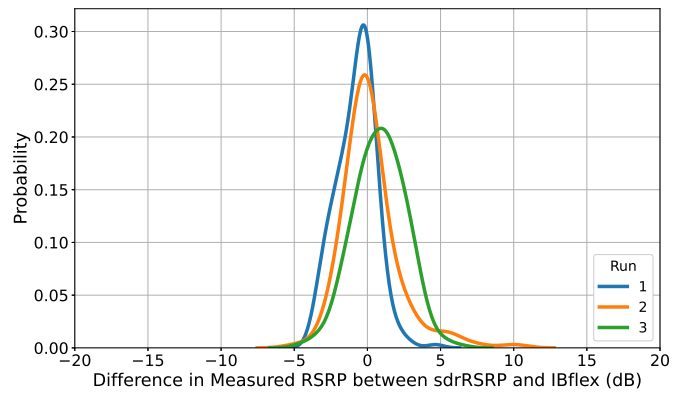

Fig. 3. The Probability Density Functions (PDF) of the discrepancies between the `sdrRSRP` and the IBflex for the three measurement runs.

To calibrate an `sdrRSRP` to a phone's RSRP measurements, we collected simultaneous moving measurements with the phone and `sdrRSRP`. Using measurements from both devices, we calculated a calibration factor to minimize the discrepancy. Again, a separate calibration factor is needed for every center frequency. We found phone calibration to be within ± 5 dB of the high-quality calibration, an improvement over the ± 20 dB uncertainty of an uncalibrated SDR.

C. Evaluation

To evaluate the accuracy of `sdrRSRP`, we conducted three moving measurement runs while simultaneously recording with `sdrRSRP` and PCTEL's IBflex [22].

The IBflex was used to measure ground truth RSRP, a calibrated drive test device primarily used by mobile network operators to monitor network performance. It has an absolute accuracy of ± 1 dB. When configured to measure all bands of a particular network, the IBflex generates an RSRP measurement approximately every 0.4 seconds.

We ran `sdrRSRP` on an Ettus USRP B210 and an Acer Predator Helios 300 laptop. The SDR had a GPS disciplined oscillator (Jackson Labs TCXO) to provide timing and location, and was configured with a static 50 dB receiver gain.

During the three runs, the IBflex and `sdrRSRP` recorded simultaneously with a shared antenna (PCTEL OP451) through a power splitter. `sdrRSRP` recorded in *Subframe RSRP Mode*. Each run covered 2 kilometers in downtown Boulder, Colorado. We limited the evaluation of the measurements to the RSRP from the strongest cell tower and within the trusted power limits, as shown in Table II. Run 1 measured signals from seven T-Mobile towers on band 2 (1957.5 MHz). Run 2 measured signals from seven T-Mobile towers on band 41 (2525.8 MHz). Run 3 measured signals from five AT&T towers on band 66 (2137.5 MHz). Runs 1 and 3 measured FDD towers, and run 2 measured TDD towers.

After the runs, we performed a rolling average across the measurements from both devices. Figure 3 shows the distribution of differences between the devices for the three runs. First, the mean difference between the devices is 0.2

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±0.3 dB standard error (95% confidence). This confirms that the sdrRSRP calibration procedure described in Section IV-B holds true across multiple runs. The standard deviation of the discrepancies between the devices is 1.8 dB. Using quadratic addition to propagate the uncertainty of the IBflex, the resulting accuracy of the calibrated sdrRSRP is ±3.5 dB (95% confidence). Finally, the Mean Absolute Error (MAE) over the three runs is 1.4 dB, indicating the close agreement between the devices.

D. Advantages and Limitations

The ability of sdrRSRP to continuously measure reference signals makes it well-suited for measurement campaigns evaluating propagation in vehicular scenarios. At driving speeds, e.g., 100 km/h, devices sampling once per second would collect data only every 30 meters, which is impractical for investigating shadowing and other large scale fading effects in high-speed rail and highway environments. In contrast, continuous measurement enables arbitrary distance resolution, underscoring an advantage of an open-source system like sdrRSRP.

A limitation of many low-cost SDRs is the trade-off between dynamic range and error introduced by using the variable gain amplifier within the device [25]. The B210, for instance, has an internal receive amplifier that can adjust during operation to extend the power measurement range. Unfortunately, we found that the effects of the variable gain amplifier are not always linear (e.g., a 20 dB gain adjustment does not provide exactly 20 dB of realized gain). To avoid determining a calibration factor for every possible gain setting, we calibrated for three receive (Rx) gains.

Table II shows the three Rx gain settings and the minimum and maximum RSRP measurements that can be trusted at that gain. We determined the minimum as 10 dB above the sdrRSRP system's noise floor, while the maximum corresponds to the 1-dB compression point.

These values were experimentally determined in a laboratory setting. To measure the noise floor of an RSRP measurement at a fixed Rx gain, the sdrRSRP was terminated with a 50Ω resistive load and set to capture samples at 11.52 MS/s. After recording for five seconds, the I/Q was processed with the same RSRP algorithm described in Section II.

To determine the 1-dB compression point, we used a second SDR running srsENB as a tower, and connected both sdrRSRP and the IBflex with a power splitter. The tower's transmit power was slowly increased via a hardware attenuator. The 1-dB compression point was identified where sdrRSRP differed by more than 1 dB from the IBflex measurement.

V. RSRP vs. CW Propagation

In this section, we compare RSRP measurements to CW propagation measurements. Our specific goal in evaluating this is to determine if RSRP can be effectively used to produce and validate propagation models previously informed by CW measurements. After transmitting 4G and CW waveforms together from a rooftop, we performed moving propagation measurements of the signals.

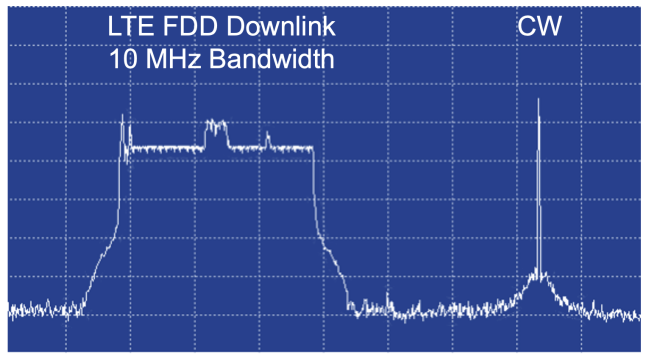

Fig. 4. A field measurement performed by a signal analyzer of the LTE downlink and CW signals in the CW and RSRP comparison measurements.

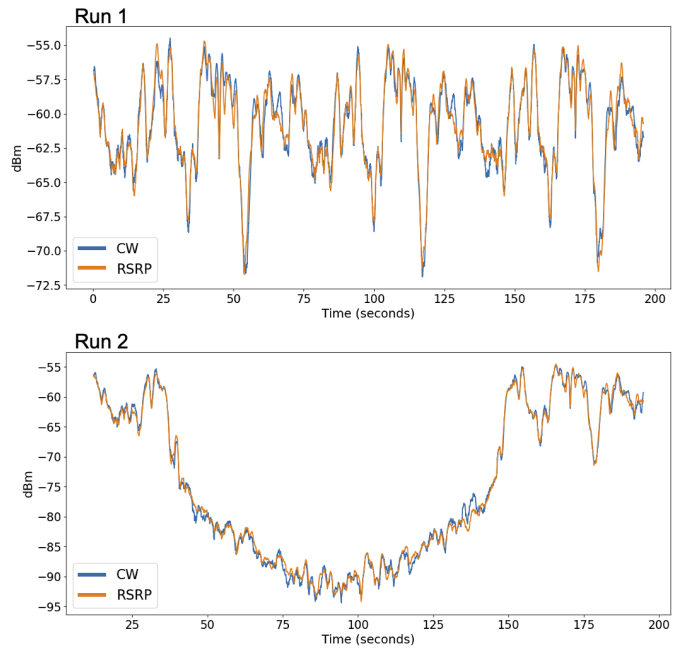

Fig. 5. The smoothed CW and smoothed RSRP measurements plotted together for runs 1 and 2.

A. Measurement Setup

For this study, we transmitted the 4G and CW waveforms seen in Figure 4. We used srsENB to generate the 4G signal. The center frequency was 3470 MHz, with a 10 MHz bandwidth. During the measurements, the cell tower was in a fully active state, and we ensured it had no UEs attached. The power seen in the 4G signal in Figure 4 comes strictly from reference signals, synchronization signals, and other broadcast information.

We obtained a Special Temporary Authority license from NTIA to temporarily transmit these waveforms at the specified frequencies, power levels, and locations. sdrRSRP was used to measure the 4G signal, which was transmitted in an unoccupied band to prevent interference. As a result, neither the IBflex nor other laboratory equipment could measure the signal, as it fell outside configurable bands. This experiment would not have been possible without sdrRSRP, which offers

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the flexibility to measure at any frequency and enables the continuous measurement required for simultaneous comparison studies.

For the CW signal, we used the Keysight Signal Generator with a center frequency of 3485 MHz. This left a 10 MHz wide buffer between the signals. Lab testing showed no measurable impact from one signal to another. The signals were joined with a power combiner and transmitted from the same antenna. Measured with a power meter at the input to the combiner, each signal was 13 dBm. The transmit antenna was located 48 feet above the ground on the edge of a roof.

For the receiver, we used a Keysight PXA to measure the CW and the `sdrRSRP` to measure the RSRP. The PXA is a calibrated, lab-grade, spectrum analyzer. The measurement devices shared an antenna.

We performed two measurement routes. Run 1 followed a loop three times around a parking lot, was predominantly within line-of-sight (LOS), and stayed between 70 and 100 meters from the transmitter. Run 2 was an out-and-back route, in predominantly non-line-of-sight (NLOS) due to building obstructions. The route reached a maximum of 300 meters from the transmitter. We maintained 15 km/h during the runs.

B. Evaluation

We performed a (60-wavelength [26]) rolling average to remove the fast-fading affects in the measurements of both waveforms. Figure 5 shows the smoothed data from both runs. We compared the discrepancies between both waveforms and the MAE for each run was 0.5 dB and 0.7 dB, respectively. We see no variation in accuracy when measuring in LOS versus NLOS or when measuring at the highest powers of -55 dBm versus the lowest powers of -95 dBm.

The narrowband CW measurement technique is commonly used by RF propagation engineers to provide data for propagation modeling efforts. This is the first study to demonstrate that the wideband RSRP technique yield measurements with negligible differences (< 1 dB), and the first study to validate the use of all RSRP-measuring devices, including smartphones and laboratory equipment, for propagation measurements.

VI. Conclusion

In this study, we introduce `sdrRSRP`, an open-source SDR toolset that offers comprehensive, low-level control over CRS measurements. Through extensive laboratory evaluation and a real-world measurement campaign, we showed that a properly calibrated SDR can deliver CRS power measurements with an accuracy of ± 3.5 dB with a MAE of 1.4 dB. Additionally, we compared the wideband RSRP measurements technique to the narrowband CW measurement technique commonly used by RF propagation engineers. Our experimental results demonstrate that the two techniques yield measurements with negligible differences of < 1 dB.

This study validates the hypothesis that the reference signals transmitted from every 4G tower present an enormous signal-of-opportunity for the widespread collection of channel propagation measurements. Propagation measurements from reference signals could offer much-needed data to meet the demands of new neural network- and ray tracing–based propagation models. They also offer measurements in RF channel path geometries and frequencies relevant to cellular networks, terrestrial broadcast, and short-range indoor and outdoor communications.

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